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Information Structure

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Information Status and Referentiality in Discourse: A Case Study of Coreference in Dan Barber's TED Talk on Sustainable Fishing

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1 Introduction

The question of how language encodes information is never neutral, and the analysis of discourse offers a way of uncovering the strategies by which speakers guide their audience through complex topics. In recent decades, research in pragmatics has paid increasing attention to information status, reference, and the ways in which linguistic forms correspond to the cognitive accessibility of discourse entities. The model developed by Gundel, Hedberg, and Zacharski in their influential 1993 article, *Cognitive Status and the Form of Referring Expressions in Discourse*, offers a particularly useful framework. Their "Givenness Hierarchy" proposes a systematic relationship between the cognitive status of a referent (whether it is "in focus", "activated", "familiar", "uniquely identifiable", "referential", or merely "type identifiable") and the choice of linguistic form, from pronouns to definite descriptions to indefinites (Gundel et al. 275). This theory provides the foundation for many contemporary studies or reference and remains a benchmark for applied discourse analysis.

The present paper takes this framework as its starting point and applies it to a text that, on the surface, may appear far removed from theoretical linguistics: a TED talk on sustainable fishing delivered by chef Dan Barber. Yet Barber's talk is an ideal case study precisely because it is a persuasive and narrative-driven discourse, balancing personal anecdotes, scientific explanation, and rhetorical appeals. At roughly sixteen minutes in length, the talk contains enough linguistic material for systematic annotation, while also being constrained enough to allow for detailed, qualitative analysis. By annotating the talk at the word level for coreference chains and information status, the paper seeks to trace how Barber introduces, maintains, and shifts reference across his narrative and explanatory passages.

The choice of text is not incidental. Sustainable fishing is a matter of urgent global concern, situated at the intersection of ecological science, agricultural practice, and consumer culture. Barber's talk draws attention to alternative models of fish farming, focusing on the case of Veta La Palma in southern Spain. His description of the farm, the surrounding ecosystem, and the culinary and environmental implications of its practices provides a rich dataset in which various types of noun phrases occur: personal pronouns (*I, he, we*), demonstratives (*this,*

that), generics (*you know, you want on a fish farm*), and definite descriptions (*the fish, the flamingos, the system*). The annotation of these expressions allows us to see how reference functions not merely as a mechanism for identifying entities, but as a rhetorical tool for framing success, responsibility, and ecological interconnectedness.

The paper's central research questions can therefore be framed as follows: first, how are referential expressions distributed across the talk, and which NP types are especially significant for the discourse? Second, how do these expressions map onto the categories of the Givenness Hierarchy, and what patterns emerge in the use of "in focus", "activated", or "referential" entities? Third, what do these patterns reveal about the relationship between narrative discourse and expository discourse? Finally, what implications can be drawn from this analysis for our understanding of persuasive sustainability rhetoric?

In addressing these questions, the paper combines qualitative and quantitative methods. The word-level annotation file contains each token of the TED talk, with information on syntactic role, NP type, coreference ID, and information status category. From this data, examples will be selected to illustrate key contrasts, such as the difference between situational pronouns (*I, you*) and discourse-deictic demonstratives (*that was fascinating*), or between generic expressions (*a farm, you want …*) and specific definite descriptions (*the fish, the system*). At the same time, quantitative counts of information status categories will provide an overview of how often Barber keeps entities "in focus", how frequently he reactivates them, and how many new referents are introduced at different points in the talk.

The significance of this analysis lies in the way it bridges linguistic theory and applied discourse. By bringing Gundel et al.'s hierarchy into dialogue with a contemporary text on sustainability, the paper demonstrates that theoretical models of reference are not abstract exercises but tools for understanding how real speakers persuade audiences and shape perceptions of environmental issues. Moreover, the case study shows how ecological discourse itself relies on careful manipulation of information structure: Barber's talk succeeds rhetorically because it not only tells a story about fish farming, but also continually calibrates the audience's cognitive focus, directing attention to certain entities (the farm, the flamingos, the system)

while backgrounding others. In doing so, he demonstrates the very dynamics the Givenness Hierarchy describes.

2 Theoretical Framework

Any systematic study of reference in discourse requires a theoretical basis that can account for the speaker's choice of form and the hearer's interpretation of meaning. The framework adopted in this paper is the Givenness Hierarchy proposed by Gundel, Hedberg, and Zacharski in 1993, supplemented by the annotation guidelines for coreference and information status used in the present analysis. Together these sources provide both the conceptual foundation and the practical procedures for assigning values to the tokens in Barber's TED talk. Earlier work had already emphasized the importance of distinguishing given and new information in discourse. Prince (1981) developed a taxonomy of given–new categories, showing that speakers systematically signal whether information is assumed to be known, accessible, or entirely new to the hearer. Ariel (1990) similarly proposed an accessibility scale, ranging from full names and definite descriptions to pronouns, where the choice of form reflects assumptions about the hearer's mental accessibility of the referent. These approaches provide the foundation on which Gundel et al. refined their more detailed hierarchy.

2.1. The Givenness Hierarchy

The Givenness Hierarchy was introduced as a way of mapping forms of referring expressions to cognitive statuses in the hearer's mind. Gundel et al. argue that a speaker's choice between pronouns, demonstratives, definite descriptions, and indefinites is not arbitrary but reflects assumptions about what the hearer can identify, retrieve, or infer at a given moment. The hierarchy consists of six ordered statuses: in focus, activated, familiar, uniquely identifiable, referential, and type identifiable (Gundel et al. 275). Each status is associated with a set of forms: for example, unstressed personal pronouns presuppose that the referent is "in focus", while demonstratives correspond to "activated", and definite descriptions indicate "uniquely identifiable".

Crucially, the hierarchy is implicational. If an entity is in focus, it is by definition also activated, familiar, and so on down the hierarchy. Conversely, if an

entity is only type identifiable, no stronger assumptions can be made about its status. This implicational nature allows the model to account for both the economy and flexibility of natural language reference. Speakers economize by choosing the minimal form necessary, while hearers use the hierarchy to recover intended referents. From a cognitive perspective, this principle corresponds to what Chafe (1994) describes as the management of consciousness in discourse, where attention shifts dynamically between what is active, accessible, or backgrounded. The categories of the hierarchy thus align closely with states of consciousness that listeners track during spoken interaction.

For the purposes of this study, the Givenness Hierarchy provides the conceptual scaffolding. When Dan Barber says, "I called him", the choice of pronoun presupposes that *I* (the speaker) is situationally given and that *him* (Miguel) is activated within the discourse. When he says, "the system", the definite NP presupposes that the system is uniquely identifiable to the hearer, even if it is newly introduced. Thus, the hierarchy offers a principled way of interpreting such choices.

2.2. Coreference Annotation Guidelines

While the hierarchy provides the theoretical background, annotation requires explicit rules. The Coreference Guidelines used here define the procedure for identifying noun phrases (NPs), assigning them a type, and linking them into coreference chains. NP types include pronouns (personal, demonstrative, possessive), definite NPs, indefinite NPs, and proper names. Each NP is the coded for REF (reference status) as NEW, OLD, GEN (generic), EXPL, (expletive), IDIOM (idiomatic), or BOUND (possessive dependent).

The guidelines specify that each discourse entity must be assigned a COREF ID. For instance, all occurrences of *Miguel* and the pronouns *he, him, his* referring to him share a COREF ID. Similarly, *the fish, our fish*, and *they* (when referring to the Spanish company's fish) belong to one chain. These chains make it possible to trace how referents are introduced, maintained, and reactivated.

Special categories prevent over-annotation. EXPL is used for non-referential *it* (*It's raining, It turns out*). IDIOM applies to fixed expressions like *you know* or *by the way*, where pronouns do not function referentially. GEN captures generic reference, as in *a fish farm* or *you want on a farm*. BOUND covers possessives that

depend on another NP (*his head, its predators*). These distinctions ensure that only genuine discourse referents are tracked.

2.3. Information Status Guidelines

The Information Status Guidelines provide the operational link between Gundel's hierarchy and annotation practice. Each NP is assigned one of the IS values: IN FOCUS, ACTIVATED, UNIQUELY IDENTIFIABLE, REFERENTIAL, TYPE IDENTIFIABLE. These correspond to the cognitive statuses of the hierarchy but are applied more concretely to the tokens in the transcript.

The guidelines define these categories in functional terms. IN FOCUS applies to subject and topical entities that are currently at the centre of attention, such as *I* (Dan Barber) or *he* (Miguel) in narrative clauses. ACTIVATED is used for entities that are salient but not topical, typically objects or previously mentioned participants, like *him* or *the system* when not in subject position. UNIQUELY IDENTIFIABLE is used for definites that introduce specific entities, such as *the night before* or *the head of the company*. REFERENTIAL applies to indefinites that denote particular but previously unmentioned entities, like *a perfect environment*. TYPE IDENTIFIABLE covers bare indefinites that establish only a category, as in *a fish farm*.

The relation between REF and IS is crucial. An NP may be REF=NEW but IS=UNIQUELY IDENTIFIABLE if it is a definite introduced for the first time. Likewise, a REF=OLD pronoun may be IS=ACTIVATED if it appears in object position. This separation of reference and information status allows for nuanced analysis. The guidelines thus operationalize the hierarchy by giving precise instructions for coders.

2.4. Relevance to the Case Study

Together, the Givenness Hierarchy and the annotation guidelines provide a comprehensive framework. The hierarchy explains why different forms exist and the guidelines explain how to assign categories in practice. Applying both to Barber's TED talk makes it possible to investigate how a persuasive, narrative text manages reference. The talk offer rich examples of each category: situational

pronouns (*I, you*), discourse-deictic demonstratives (*that was fascinating*), generic reference (*a farm*), idioms (*you know*), and specific definites (*the flamingos, the system*). Each of these maps onto the IS categories in ways predicted by Gundel et al.

This framework also highlights the rhetorical work of information structure. Barber keeps Miguel and the farm in focus across extended stretches, while reactivating entities like *the fish* or *the system* when needed. He introduces new entities carefully, often with vivid descriptions (*a literal pink carpet*), making them referentially salient. He uses generic forms to generalize about farming practices (*a farm that doesn't feed its animals*), while pronouns maintain cohesion in personal anecdotes. By applying the hierarchy and guidelines systematically, the analysis reveals how reference is managed both to maintain narrative flow and to support the persuasive message about sustainability.

3 Methodology

The empirical basis of this study is a full transcription of Dan Barber's TED talk on sustainable fishing, which runs approximately sixteen minutes. The transcript was segmented at the word level, with each token occupying one row of an Excel spreadsheet. This design allowed for fine-grained annotation and ensured that no referential expression would be overlooked. In total, every token in the transcript was considered, though not every token received a value: punctuation, function words without referential content, and idiomatic fillers were left blank where appropriate, in accordance with the annotation guidelines.

The spreadsheet contained multiple columns corresponding to the annotation categories. The first of these was NP_TYPE, which identified the form of the noun phrase or pronoun. Categories included personal pronouns (pron.pper), demonstrative pronouns (pron.pds), possessive pronouns (pron.ppos), definite noun phrases (def-np), indefinite noun phrases (indef-np), and proper names. This column served as the initial classification step, distinguishing between forms whose distribution is central to Gundel et al.'s Givenness Hierarchy and those that play a marginal or idiomatic role.

The second column, COREF, contained the identifiers for discourse entities. Each entity was assigned a label, and all subsequent references to that entity received the same COREF value. For instance, *Miguel, he, him,* and *his* were consistently annotated with the same identifier. Similarly, *the fish, our fish*, and *they* (when referring to the Spanish company's stock) were linked as one chain, while generic uses of *fish* were excluded or assigned a different identifier. These COREF chains allowed for the tracking of referents across the narrative, making it possible to analyse how entities were introduced, maintained, and reactivated.

The third column, REF, marked the referential status of each NP. Following the guidelines, six categories were used: NEW, OLD, GEN (generic), EXPL (expletive), IDIOM (idiomatic), and BOUND (possessive-dependent). NEW was applied when an entity was introduced into the discourse for the first time, as in *a fish farm* or *a literal pink carpet*. OLD marked entities that had already been introduced, as with repeated mentions of *Miguel* or *the system*. GEN indicated generic reference, such as *you want on a farm*. EXPL captured non-referential *it* in constructions like *It turns out*. IDIOM was used for fixed expressions, notably *you know* or *by the way*. Finally, BOUND applied to possessives that derive their reference from another NP, as in *his head* or *its predators*. This level of annotation thus distinguished between true discourse referents and expressions that did not add to the set of entities under discussion.

The fourth column, IS, encoded the information status of each NP. Here the categories derived directly from Gundel et al.'s Givenness Hierarchy: IN FOCUS, ACTIVATED, UNIQUELY IDENTIFIABLE, REFERENTIAL, and TYPE IDENTIFIABLE. Each NP was annotated according to its cognitive status in the discourse. IN FOCUS was reserved for entities currently at the centre of attention, often realized as subjects, such as *I* in Barber's narration or *he* when Miguel is the subject of the reported action. ACTIVATED referred to entities that were salient but not topical, typically objects or pronouns in non-subject positions, such as *him* or *it*. UNIQUELY IDENTIFIABLE was assigned to definites that presuppose uniqueness, like *the night before* or *the system*. REFERENTIAL covered indefinites that denoted a specific but previously unmentioned entity, such as *a perfect environment*. TYPE IDENTIFIABLE was used for bare indefinites that establish only a category, as in *a farm* or *a set change*.

The assignment of REF and IS categories followed a two-step logic. First, REF determined whether the entity was new, old, generic, or non-referential. Then IS was used to capture its cognitive status. The same NP could therefore receive different combinations, depending on its discourse role. For example, *the fish* might be REF=OLD but IS=IN FOCUS when functioning as subject, or REF=OLD but IS=ACTIVATED when functioning as object. This separation between reference and information status reflects the theoretical distinction emphasized by Gundel et al. and operationalized in the guidelines.

All annotation was conducted manually in Excel. No additional software or automated tools were used. This choice ensured maximum control and consistency, though it also placed limits on the scale of the analysis. The annotation was performed by a single coder, the author of this paper, working systematically through the transcript row by row. While this means that no inter-annotator reliability could be calculated, consistency was maintained by repeated reference to the guidelines and by iterative review of earlier decisions. In practice, coding challenges arose especially with ambiguous cases, such as discourse-deictic *that*, generic *you*, or idiomatic uses of pronouns. These were resolved by consulting the examples in the guidelines and by applying the principle of excluding non-referential tokens from the coreference chains.

The methodological choice to annotate every token, rather than only the noun phrases, was deliberate. It allowed for a comprehensive mapping of the discourse and prevented the overlooking of subtle cases. For instance, idiomatic expression such as *you know* or *by the way* were clearly identifiable as IDIOM in REF and left blank in IS, ensuring that the dataset distinguished between true referential expressions and pragmatic markers. Similarly, expletive *it* in *It turns out* was consistently marked as EXPL, preventing false links in the coreference chains.

Finally, the dataset produced by this annotation forms the empirical foundation for the analysis to follow. By capturing both the form (NP_TYPE), the referential status (REF), the discourse continuity (COREF), and the cognitive accessibility (IS) of each NP, the spreadsheet provides a detailed account of how Barber manages reference. The methodology thus operationalizes the theoretical framework and produces a dataset from which both qualitative examples and quantitative patterns

can be extracted. The latter will be discussed in the Analysis section, while the present account establishes the reliability and coherence of the annotation process itself.

For reasons of space, only a short excerpt from the annotation is reproduced in Appendix A, illustrating the structure of the dataset and typical coding decisions. The full annotated transcript is included as Appendix B in the digital submission. This ensures that the complete dataset remains accessible for verification while keeping the printed thesis concise.

4 Analysis

The annotated transcript of Dan Barber's TED talk provides a detailed view of how referential expressions are used in a persuasive narrative about sustainable fishing. This section examines the results of the annotation, with attention to the distribution of noun phrase types, reference categories, and information status values. The analysis is organized into four subsections: NP types of interest, reference status distribution, information status distribution, and the interaction of REF and IS. Each subsection combines quantitative overview with qualitative illustration, drawing examples directly from the text.

4.1. NP Types of Interest

The first step in analysing Barber's TED talk was to identify and classify the types of noun phrases (NPs) present in the transcript. Doing so provides a foundation for understanding how reference and information status are distributed across the text. The spreadsheet annotation distinguishes categories such as pronouns, definites, indefinites, proper names, demonstratives, and idiomatic expressions. The distribution already shows a marked division between Barber's narrative mode, where pronouns and proper names dominate, and his expository mode, where definites and indefinites become more prominent. This alternation reflects the hybrid character of the TED talk as both a personal anecdote and a persuasive ecological argument.

Among NP types, personal pronouns are by far the most frequent, with over 250 occurrences. These include first-person (*I, me, my, we, us*), second-person (*you*), and third-person (*he, him, his, she*). Their dominance is unsurprising in a spoken narrative, but their rhetorical function is worth emphasizing. When Barber says, "I didn't know anything about fis*h*" the pronoun *I* places him in the role of a learner, aligning his persona with the audience rather than with the experts he interviews. Elsewhere he shifts to generic reference: "For the past fifty years, we've been fishing the seas like we clear-cut forests". Here, *we* is annotated GEN, presenting a broad collective of humanity responsible for overfishing, rather than Barber and his immediate audience. By contrast, situational (SIT) *we* surfaces in moments of advice, such as "We should call it what it is", where speaker and audience are explicitly aligned. The alternation between GEN and SIT *we* illustrates Barber's rhetorical skill in moving between universalising claims and calls to immediate collective action.

Indefinite noun phrases form another significant class, appearing around 150 times in the transcript. They often introduce new referents in vivid rhetorical contexts. "A drop in the ocean", "a Whitney Houston song", or "a business in liquidation" are not simply referents but metaphors that capture Barber's evaluative view. Such indefinites are crucial for creating imagery that engages listeners. They also contrast bare plural indefinites such as *fish, farmers,* or *predators*, which are used to generalize across kinds. The annotation marked these as TYPE IDENTIFIABLE or REFERENTIAL depending on whether they represent categories or specific instances. In both cases, Barber uses indefinites to balance specificity with generality, offering both illustrative anecdotes and broad ecological claims.

Definite noun phrases occur just over one hundred times, with forms such as *the system, the farm*, or *the future of good food*. These definites are often introduced after the entity has been made significant, either through narration or interpretation. For example, after describing Miguel's ecological practices, Barber can refer back to *the system* without further introduction, assuming the audience can uniquely identify what is meant. Definite possessives such as *its predators* or *his head* show how definiteness combines with dependency as the farm's predators are not new discourse referents, but entities understood through their relation to the already

established farm or system. These definites play an expository role, allowing Barber to move from anecdote to general claim.

Proper names are relatively rare, with fewer than forty occurrences, but they play a strategic role. The central figure is Miguel, the Spanish chef-biologist whose farm becomes the core case study. Other names, such as *Charles Darwin* or *Crocodile Dundee* appear in metaphors: "He's like three parts Charles Darwin and one part Crocodile Dundee". These names do not establish new discourse referents in the same way as Miguel but function metaphorically, shaping how the audience conceptualizes him. Their annotation as NEW referents reflect their first mention, but in discourse terms they serve as stylistic additions rather than ongoing entities.

Demonstratives like *this, that, these, those* appear both as pronouns and determiners, with around thirty instances. They perform crucial discourse-deictic work, often summarizing or commenting on prior stretches of talk. For instance, "That's success" evaluates the preceding description of ecological balance, while "this love story" labels the narrative Barber has spun about the farm. These demonstratives move beyond simple reference to embody Barber's evaluative position, making them especially important in a persuasive context.

Finally, idiomatic or non-referential expressions deserve mention. These were annotated as IDIOM or EXPL, ensuring that they did not interfere with the referential chains. Instances such as *you know* and *by the way* were treated as idiomatic discourse markers rather than as genuine referents, while expletive *it* in *it turns out* was classified as EXPL. Although relatively infrequent, these forms play an important role in structuring the flow of the talk, signalling shared understanding, transitions, or rhetorical surprise. Excluding them from the referential inventory prevents distortions in the quantitative profile and highlights that not all pronouns and determiners contribute equally to discourse reference.

Taken together, the distribution of NP types shows how Barber orchestrates his argument by combining different resources. Pronouns and proper names sustain the narrative core, indefinites provide vivid imagery, definites stabilize the expository layer, demonstratives offer evaluative framing, and idiomatic expressions manage discourse flow without introducing new entities. This layering reflects the dual

character of the TED talk as it is a personal story of discovery on the one hand and a persuasive ecological argument on the other.

4.2. Reference Status (REF) Distribution

Each NP was further annotated for reference status (REF), distinguishing between OLD, NEW, GEN, SIT, BOUND, IDIOM, and EXPL. This classification captures whether an entity had already been introduced, was newly established in discourse, was generic, tied to immediate speech situation, structurally dependent, idiomatic or expletive. The distribution of these categories in the dataset was as follows: OLD (271 cases), NEW (140), GEN (28), SIT (63), BOUND (23), IDIOM (11), EXPL (2).

OLD was by far the most frequent category. Once entities such as *Miguel, the system,* or *the farm* had been introduced, they were reactivated through pronouns or definite descriptions. This accounts for the high number of OLD tokens and underscores the cohesion of the talk, which relies on a small cast of recurring participants. For instance, *he* and *him* consistently refer back to Miguel, while *it* and *its* often continue reference to *the system*. These OLD annotations show that continuity is achieved not by increasing new referents but by maintaining accessibility of the key ones.

NEW referents, while less numerous, were rhetorically striking. They included indefinite metaphors such as “a drop in the ocean” or “a business in liquidation”, as well as proper names like *Charles Darwin*. Such NEW entities were annotated only at their first mention and often did not persist in the discourse. Their function is primarily to develop an evaluative point or provide a memorable image. This explains why the number of NEW tokens is smaller than OLD, but their persuasive impact is disproportionately large.

GEN captured generalised reference, most often with pronouns. Examples include “Isn’t a thriving bird population the last thing you want on a fish farm?” where *you* is not addressed to the immediate audience but represents a generic agent, and “For the past fifty years, we’ve been fishing the seas…”, where *we* indicates humanity as a whole. GEN annotations reveal how Barber universalises his claims, shifting from narrative particulars to collective responsibility.

SIT was reserved for references anchored in the speech event. This included first-person *I* for the speaker, inclusive *we* in exhortations such as "We should call it what it is" and second-person *you* in direct appeals like "… for you and me as well". SIT reference directly involves the audience, making them co-participants in Barber's evaluations and calls to action. These annotations demonstrate how Barber balances general claims with situationally grounded engagement.

BOUND accounted for structurally dependent possessives, such as *its predators* or *his head*. These were annotated as BOUND because they do not function as independent discourse referents but derive their meaning from their relation to an established NP.

IDIOM and EXPL were used for forms that do not contribute to reference. "you know" and "by the way" were annotated as IDIOM, and expletive *it* in "it turns out" as EXPL. These annotations ensured that such expressions were excluded from the referential chains, keeping the analysis focused on genuine entities.

Overall, the REF distribution highlights a discourse strategy built on continuity rather than expansion. OLD referents maintain cohesion, NEW ones punctuate the argument with vivid imagery, GEN and SIT allow Barber to vary the scale of address, and BOUND, IDIOM and EXPL remain peripheral.

4.3. Information Status (IS) Distribution

In addition to reference status, each NP was annotated for information status (IS) according to the categories defined by the Givenness Hierarchy. These include IN FOCUS, ACTIVATED, UNIQUELY IDENTIFIABLE, REFERENTIAL, FAMILIAR, and TYPE IDENTIFIABLE. The distribution in the dataset was as follows: IN FOCUS (240), ACTIVATED (88), UNIQUELY IDENTIFIABLE (62), REFERENTIAL (57), FAMILIAR (23), TYPE IDENTIFIABLE (4). The results confirm that Barber's discourse relies heavily on a narrow set of highly accessible referents, while other statuses are used more selectively.

IN FOCUS was the most frequent category, reflecting the importance of Barber himself and Miguel in the talk. Pronouns such as *I* and *he* were often maintained as subjects across stretches of narration, keeping them continuously in focus. For example, in "He looked at me … I had just quoted …," the subjects *he* and *I* remain

IN FOCUS throughout, guiding the listener's attention and sustaining narrative coherence. This dominance of IN FOCUS shows that Barber constructs the talk around a small set of protagonists.

ACTIVATED marked entities that were relevant but not currently topical. These were often objects or previously mentioned entities that were reintroduced in non-subject positions. For instance, in "He looked at me," *me* was annotated as ACTIVATED, since the referent is accessible but not the discourse focus. Similarly, pronouns such as *it* referring back to metaphoric indefinites (e.g. "a drop in the ocean") were often ACTIVATED. This category indicates that Barber keeps a short list of salient alternatives active in the discourse without shifting the primary focus away from the central figures.

UNIQUELY IDENTIFIABLE applied primarily to definite noun phrases that presupposed a unique referent retrievable from the discourse or situational context. Expressions such as *the system*, *the farm*, or *the health of its predators* were annotated as UNIQUELY IDENTIFIABLE once the context allowed them to be identified without ambiguity. This demonstrates how Barber uses definites to move from narrative description to expository generalization, relying on the assumption that the audience can uniquely recover the intended referent.

REFERENTIAL was assigned to indefinites that introduced new, specific referents into the discourse. Examples include "a recipe for the future" or "a business in liquidation." These NPs do not represent kinds but concrete instances or images that Barber uses to drive home evaluative points. Their relatively frequent use (57 cases) shows that Barber punctuates his talk with striking new referents at rhetorical peaks.

FAMILIAR and TYPE IDENTIFIABLE were rare but significant. *FAMILIAR* captured referents that presuppose shared cultural knowledge or that had already been labelled in the discourse, such as "this love story." *TYPE IDENTIFIABLE* applied to kind-level references such as *beef cattle* or bare plural *birds*, which support general claims rather than concrete discourse participants. Although numerically small, these categories highlight how Barber occasionally situates his argument in broader cultural or categorical frames.

Taken together, the IS distribution demonstrates that Barber's discourse structure is anchored by IN FOCUS referents, primarily himself and Miguel, while ACTIVATED entities and UNIQUELY IDENTIFIABLE definites expand the network without destabilizing it. REFERENTIAL indefinites add rhetorical force, and the minor categories FAMILIAR and TYPE IDENTIFIABLE situate the talk in wider conceptual contexts. The annotation thus confirms the predictions of the Givenness Hierarchy as it forms a map systematically onto levels of cognitive accessibility, and Barber exploits this mapping to balance narrative immediacy with argumentative range.

4.4. Interaction of REF and IS

The interaction of reference status (REF) and information status (IS) provides the clearest picture of how accessibility is managed across the talk. Rather than operating in isolation, the two systems combine to yield characteristic pairings that shape narrative continuity, rhetorical emphasis, and audience engagement.

The most frequent and predictable combination was OLD + IN FOCUS, which formed the backbone of the narrative. Once *Miguel* and *Barber* were established, subsequent pronouns such as *he*, *him*, *I*, and *me* maintained them as OLD referents with IN FOCUS or ACTIVATED status, depending on syntactic role. This pairing explains the cohesiveness of long stretches of narration: reference to the same entities is repeated while their cognitive accessibility remains at the highest level.

A second important combination was NEW + REFERENTIAL, which signalled the introduction of vivid metaphors and images. Phrases like "a business in liquidation" or "a drop in the ocean" entered the discourse as NEW referents, annotated with REFERENTIAL IS, since they introduced concrete conceptualizations at critical points. These referents rarely stayed as OLD but functioned as striking one-off images that shaped audience interpretation.

A related but distinct pattern was NEW + UNIQUELY IDENTIFIABLE. This occurred with definite expressions that were mentioned for the first time but were nonetheless assumed to be uniquely recoverable, such as *the head of the company* or *the health of its predators*. These entities were annotated as NEW in terms of REF, since they had not been explicitly introduced, but as UNIQUELY IDENTIFIABLE in terms of IS, because the discourse provided sufficient cues for

unique identification. This combination illustrates how Barber draws on the inferential potential of definites to expand his argument while keeping the discourse economical.

GEN + IN FOCUS appeared most often with generalized pronouns like *we* or *you* in subject position. In "For the past fifty years, we've been fishing the seas like we clear-cut forests," *we* was annotated as GEN + IN FOCUS, highlighting its generic scope while still occupying the syntactic and discourse centre. Similarly, *you* in rhetorical questions like "Isn't a thriving bird population the last thing you want on a fish farm?" was GEN + IN FOCUS, indexing a generalized agent rather than the immediate audience.

SIT + IN FOCUS marked rhetorical axles where Barber directly engaged his listeners. Expressions like "We should call it what it is" and "not just for you and me as well" were SIT in REF and IN FOCUS in IS. This combination recruited the audience into Barber's evaluative stance, creating immediacy and alignment at key argumentative junctures.

A closer passage-level analysis illustrates these dynamics. Consider the description: "A farm that doesn't feed its animals, a farm that measures its success by the health of its predators." Here, *farm* was annotated as NEW + TYPE IDENTIFIABLE, introducing a general category rather than a token; *its animals* contained a BOUND possessive, while *success* and *health* became UNIQUELY IDENTIFIABLE once the evaluative frame had been established. A few lines later, in "It's a river that carries with it all the things that rivers tend to carry these days," *it* referred back to the Guadalquivir and was annotated OLD + IN FOCUS, while *river* was a re-description supporting the same referent. *Chemical contaminants* and *pesticide runoff* were NEW + TYPE IDENTIFIABLE, broadening the expository scope.

The alternation between GEN and SIT pronouns is especially revealing. In generalized statements Barber uses GEN to universalize his point, while in exhortations he uses SIT to bind himself and the audience together. This shifting between GEN + IN FOCUS and SIT + IN FOCUS demonstrates how reference is strategically managed to scale responsibility from society at large to the people in the room.

Overall, the interaction of REF and IS shows that Barber relies on a limited set of recurrent patterns—OLD + IN FOCUS for continuity, NEW + REFERENTIAL for rhetorical images, NEW + UNIQUELY IDENTIFIABLE for expository expansion, and GEN/SIT + IN FOCUS for scaling the scope of address. These combinations illustrate how form and accessibility work together to structure persuasion.

5 Discussion

The analysis of Barber's TED talk provides strong empirical support for the predictions of Gundel et al.'s Givenness Hierarchy while also extending its application to persuasive, narrative-driven discourse. The distribution of reference types and information statuses illustrates that accessibility in discourse is not simply a matter of grammar but a resource that can be strategically managed for rhetorical effect.

First, the prominence of OLD + IN FOCUS referents shows how narrative cohesion is maintained through the repeated use of pronouns for a limited set of protagonists. This result aligns with the hierarchy's prediction that highly accessible entities are expressed with minimal forms. At the same time, Barber's alternation between GEN and SIT pronouns demonstrates how reference can be modulated to shift the scale of responsibility: GEN forms implicate humanity at large, while SIT forms directly involve speaker and audience. This alternation reveals an extension of the hierarchy into rhetorical practice, showing how accessibility distinctions can be deployed to calibrate moral responsibility.

Second, the recurrent introduction of NEW + REFERENTIAL metaphors highlights a dimension not always emphasized in theoretical treatments namely the use of vivid new referents for persuasive effect. While the hierarchy accounts for the form of indefinites, Barber's talk shows that speakers exploit the category strategically to punctuate discourse with evaluative images. These instances suggest that the "referential" category, while rare in expository texts, may be central in persuasive discourse.

Third, the findings demonstrate the complementary role of NEW + UNIQUELY IDENTIFIABLE definites. By presupposing unique identifiability, Barber is able

to introduce new concepts efficiently without overloading the audience. This practice aligns with the hierarchy's implicational logic but also illustrates how definites can operate at the boundary between NEW and OLD reference, bridging narrative description and expository generalization.

The analysis also underscores the importance of non-referential forms. Annotating idioms (you know, by the way) and expletives (it turns out) as non-referential avoided distortions in the data and highlighted their distinct discourse functions. While excluded from coreference chains, these items nonetheless structure the talk by managing transitions, signalling shared ground, and introducing evaluative turns. Their role suggests that a comprehensive account of discourse reference must also attend to forms that manage interaction rather than introduce entities.

Taken together, the findings affirm the robustness of the Givenness Hierarchy as a descriptive model while also pointing to areas where its application can be refined. The alternation of GEN and SIT pronouns, the prominence of referential metaphors, and the strategic use of definites illustrate that accessibility is not only cognitive but also rhetorical. Speakers manage reference to maintain cohesion, to generalize or particularize claims, and to align audiences with their stance.

Finally, the case study demonstrates the value of applying theoretical frameworks to contemporary ecological discourse. Barber's persuasive success rests in part on his ability to control information structure: keeping core entities in focus, activating backgrounded elements when needed, and introducing new referents through vivid images. These practices exemplify how reference is central not only to linguistic theory but also to real-world communication about urgent global issues.

6 Conclusion

This study has examined reference and information status in Dan Barber's TED talk on sustainable aquaculture, using the framework of Gundel et al.'s Givenness Hierarchy. The manual annotation of noun phrases for coreference, reference status, and information status provided both quantitative distributions and qualitative insights into how accessibility is managed in a persuasive spoken text.

The analysis demonstrated that different types of noun phrases contribute distinctively to the overall structure of the talk. Pronouns dominate in narrative passages, maintaining Barber himself, Miguel, and the ecological system in constant focus. In contrast, definites and indefinites structure expository passages: definites stabilize entities once introduced, while indefinites provide vivid metaphors that punctuate the argument with memorable images. Proper names, demonstratives, and idiomatic forms further diversify the repertoire, serving stylistic, evaluative, or discourse-organizational functions.

The distribution of reference status confirmed that continuity is secured through the repeated use of OLD referents, while NEW tokens, often metaphoric, are introduced at strategic moments. Generic and situational pronouns were shown to alternate in function: generic forms allow Barber to universalize claims about humanity's role in overfishing, while situational forms recruit the immediate audience into his evaluative stance. Possessives, idioms, and expletives remained peripheral but nonetheless important for maintaining coherence and rhetorical flow.

The information-status annotation added another dimension, revealing how accessibility is managed dynamically. IN FOCUS referents dominate, reflecting the centrality of a small set of protagonists, while ACTIVATED entities provide a surrounding frame of outstanding alternatives. Uniquely identifiable definites bridge narration and exposition by allowing Barber to introduce new but retrievable entities, and referential indefinites highlight moments of evaluative intensity. The interaction of reference and information status showed recurring patterns—such as OLD + IN FOCUS for cohesion, NEW + REFERENTIAL for metaphorical images, and NEW + UNIQUELY IDENTIFIABLE for efficient exposition—that make Barber's argument both accessible and persuasive.

Taken together, these findings highlight how reference and accessibility are not just grammatical matters but rhetorical resources. Barber's success depends on his ability to keep a compact set of entities continuously available, to punctuate the discourse with vivid new referents, and to alternate between broad societal responsibility and direct audience engagement. The analysis confirms the predictions of the Givenness Hierarchy while also showing how its categories can illuminate strategies of persuasion in contemporary ecological discourse.

Like any case study, this analysis is limited in scope. It focuses on a single TED talk, which means the findings cannot be assumed to generalize across speakers, topics, or genres without further comparative work. The reliance on manual annotation also introduces a degree of subjectivity, since coding decisions for reference and information status require interpretation even when guided by established frameworks. Finally, the spoken performance style of TED talks, with their blend of anecdote, humor, and argumentation, may foreground certain strategies that are less prominent in other communicative contexts. These limitations do not diminish the value of the findings, but they underscore the need to view them as illustrative rather than exhaustive.

Future work could extend this approach by comparing Barber's talk with other TED talks on sustainability or related themes, in order to test whether the patterns observed here—especially the prominence of metaphoric indefinites and the alternation of generic and situational pronouns—are general features of persuasive spoken discourse or specific to Barber's style. Cross-linguistic studies could further explore whether similar strategies of accessibility are deployed in other languages and cultural contexts.

In conclusion, the study has shown that reference and information structure are central to the effectiveness of Barber's TED talk. By maintaining core entities in focus, introducing new referents at strategic moments, and carefully alternating between general and situational forms, Barber exemplifies how linguistic resources for accessibility can be mobilized in the service of persuasion.

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AURIS Guidelines for Coreference Annotation. Internal manual, version 0.2.b Unpublished manuscript, 2024.

AURIS Guidelines for Information Status Annotation. Internal manual, version 0.2.b Unpublished manuscript, 2024.

Appendices

Appendix A:

1	WORD	GR	NP_TYPE	REF_AUTO	COREF	REF	IS
2546	"						
2547							
2548	# text = OK.						
2549	OK						
2550	.						
2551							
2552	# text = A farm that doesn't feed its fish, a farm that measures its success by the success of its predators.						
2553	A						
2554	farm	other	indef-np.a				
2555	that	SBJ_2					
2556	does						
2557	n't						
2558	feed						
2559	its	other_3	pron.ppos	?OLD		BOUND	
2560	fish	OBJ_2	def-np.poss	?OLD	n spanish c	OLD	ACTIVATED
2561	,						
2562	a						
2563	farm	other	indef-np.a				
2564	that	SBJ_3					
2565	measures						
2566	its	other_4	pron.ppos	?OLD		BOUND	
2567	success	OBJ_3	def-np.poss	?OLD	success	OLD	ELY_IDENTI
2568	by						
2569	the						
2570	success	other_4	def-np.poss	?OLD	success	OLD	ELY_IDENTI
2571	of						
2572	its	other_6	pron.ppos	?OLD		BOUND	
2573	predators	other_5	def-np.poss	?OLD	predators	NEW	EFERENTIA
2574	.						

Appendix B: the full excel spreadsheet is provided in digital form